

Document	Detail
Course	GIS Applications in Telecommunications
Day	Day 3 — Geo-Marketing & Customer Analytics
Version	3.0
Depends on	Day 1 (GIS fundamentals / geodatabase) and Day 2 (coverage & sites) concepts
Primary tools	Geocoding, XY Table to Point, Kernel Density, Spatial Join, Join Field, Hot Spot Analysis (Getis-Ord Gi*), Cluster & Outlier (Anselin Local Moran's I), Suitability Modeler / Weighted Overlay
Data folder	Day3_GeoMarketing/data/

1. Learning Objectives

By the end of Day 3, participants will be able to:

- Geocode customer addresses (or convert XY tables) into map points in ArcGIS Pro.
- Map subscriber distribution and concentration using Kernel Density / heat-map symbology.
- Join ARPU, churn-risk, and service-quality KPIs to geographic zones using attribute joins and Spatial Join.
- Identify high-value and at-risk customer clusters using Hot Spot Analysis and Cluster & Outlier Analysis.
- Support market-expansion decisions with Suitability Modeler / Weighted Overlay.
- Produce a short geo-marketing map package and summary table for management.

2. Day Agenda

Time	Session	Outcome
09:00–09:30	Recap Day 2 + Geo-marketing concepts	Shared vocabulary: ARPU, churn, density, hot spots

09:30–10:30	Lab 1 — Project setup, zones, customers, geocoding/XY	Subscriber points on map
10:30–10:45	Break	—
10:45–12:00	Lab 2 — Kernel Density / heat maps + KPI joins	Density surface + zone KPI layer
12:00–13:00	Lunch	—
13:00–14:30	Lab 3 — Hot Spot + Cluster & Outlier analysis	High-value / at-risk segments mapped
14:30–14:45	Break	—
14:45–15:40	Lab 4 — Market expansion suitability model	Ranked expansion zones / sites
15:40–16:00	Wrap-up, outputs checklist, quiz	Day 3 deliverables complete

3. Concepts Quick Brief

3.1 Why geo-marketing in telecom?

- Locate where subscribers actually live/work — not only where cellsites are.
- Link commercial KPIs (ARPU, churn) to geography for targeted campaigns.
- Find underserved but valuable areas for sales / network investment.
- Detect statistically significant clusters instead of guessing from raw maps.

3.2 Key terms

Term	Meaning in this course
Geocoding	Converting addresses into XY map coordinates
ARPU	Average Revenue Per User (USD/month in lab data)
Churn risk	Probability a subscriber will leave (0–1 in lab data)
Kernel Density	Creates a continuous “heat” surface of point concentration
Spatial Join	Transfers attributes based on location relationship
Hot Spot (Gi*)	Finds statistically significant high/low value clusters
Cluster & Outlier (Local Moran's I)	Finds HH/LL clusters and HL/LH outliers
Weighted Overlay / Suitability	Combines reclassified criteria with weights to rank places

4. Datasets — Synthetic (lab) + Real (optional)

Day 3 uses a complete synthetic MetroTel Capital Region package so the class can run without confidential customer data. Optional real open datasets are listed for upgrade/replacement.

4.1 Synthetic data included (use these in class)

File	Type	Records / content	Use in labs
MarketingZones.geojson	Polygon	12 marketing districts	Zone mapping, joins, suitability
MarketingZones_centroids.csv	Table	12 district centroids	Reference / labels
Subscribers_Addresses_RAW.csv	Table	600 customers with addresses + KPIs	Geocoding input style
Subscribers_ToGeocode.csv	Table	~47 unmatched address rows	Lab 1 geocode practice
Subscribers_with_XY.csv	Table	600 customers; most with LON/LAT	XY to Point, density, hot spots
Zone_KPI_Table.csv	Table	12 zone KPI summaries	Attribute Join
CellSites.csv	Points table	On-air + candidate sites	Context / suitability
CompetitorSites.csv	Points table	15 competitor sites	Competition criterion
Suitability_Weights.csv	Table	Criterion weights	Lab 4
Day3_AllTables.xlsx	Excel	All tables in sheets	Instructor quick view
REAL_DATA_SOURCES.txt	Text	Open-data links	Optional real-data path

4.2 Field dictionary — Subscribers_with_XY.csv

Field	Description	Example
CUST_ID	Unique customer ID	C00025
ADDRESS / CITY	Address text for geocoding	15 Park Ave, ...
DIST_ID / DIST_NAME	Home marketing zone	DZ-01 / CBD
PLAN_TYPE	Prepaid / Postpaid / Enterprise	Postpaid
ARPU_USD	Monthly revenue	27.40
CHURN_RISK	0–1 risk score	0.62
DROP_RATE_PCT	Service quality proxy	2.10
THROUGHPUT_MBPS	Service quality proxy	18.5
LON / LAT	WGS84 coordinates	69.16, 34.53
GEOCODE_STATUS	Matched or NeedsGeocode	Matched

4.3 Optional REAL data (replace synthetic when available)

- Administrative boundaries: HDX COD-AB for your country.
- Population grids: WorldPop or Meta High-Resolution Population Density.
- Streets/POIs: Geofabrik OSM extract for locator building / context.
- Cell/POI context: OpenCellID (training only; check license) or regulator site registers.
- Your anonymized CRM extract: CUST_ID + address or XY + ARPU + churn flags (no names/phones in class shares).

NOTE: Never use live production PII in training. If using real CRM data, anonymize IDs and remove names/MSISDNs.

5. Software & Project Setup

5.1 Required extensions / licenses

- ArcGIS Pro with privileges to run Spatial Analyst (Kernel Density, Weighted Overlay).
- Hot Spot / Cluster tools are in Spatial Statistics toolbox (standard in Pro).
- Geocoding: ArcGIS World Geocoding Service (org credits) OR local locator OR use XY fields provided.

5.2 Create the Day 3 project

- Open ArcGIS Pro → New → Map → name project MetroTel_Day3_GeoMarketing.aprx
- Set folder connection to .../Day3_GeoMarketing/data
- Insert → New Map; rename map Day3_GeoMarketing
- Right-click Map → Properties → Coordinate Systems → set to WGS 1984 Web Mercator (Auxiliary Sphere) for display OR a local UTM zone if your Day 1/2 project used UTM. Keep analysis consistent.
- Create file geodatabase: MetroTel_Day3.gdb inside the project home folder.

TIP: If Day 2 project exists, you may copy OnAir cell sites into Day 3 GDB for continuity.

5.3 Load baseline layers

- Add MarketingZones.geojson → export/import to GDB as feature class MarketingZones (Project tool if needed).
- Symbolize zones with unique values on DIST_NAME; set 40–50% transparency.
- Add CellSites.csv → Display XY Data (LON=X, LAT=Y, GCS WGS84) → export to CellSites.
- Add CompetitorSites.csv → XY to Point → CompetitorSites.
- Label CellSites with SITE_ID; use different symbols for STATUS = OnAir vs Candidate.

CHECKPOINT: You should see 12 zones, on-air sites, candidates, and competitor points.

6. Lab 1 — Geocoding & Customer Points

Objective: Place subscribers on the map from address or XY data.

6.1 Path A — Fast path using XY (recommended for timed class)

- Data → Add Data → Subscribers_with_XY.csv
- Right-click table → Display XY Data / XY Table To Point
- X Field = LON, Y Field = LAT; Coordinate System = GCS WGS 1984
- Run → output Subscribers_XY_temp; then Export Features to GDB as Subscribers
- Definition Query: GEOCODE_STATUS = 'Matched' for density tools (or delete null-geometry rows)
- Symbolology: Graduated colors / graduated symbols on ARPU_USD (5 classes, Natural Breaks)

6.2 Path B — Geocode unmatched addresses

- Add Subscribers_ToGeocode.csv
- Analysis → Tools → Geocode Addresses
- Input table = Subscribers_ToGeocode; Input Address Fields = ADDRESS (or Address + City)
- Input Locator = ArcGIS World Geocoding Service (or your local Street locator if built from OSM)
- Output = Subscribers_Geocoded
- Review Geocode Result statuses (Matched / Tied / Unmatched). Rematch interactively if needed.
- Merge / Append geocoded points with Subscribers feature class if you want one layer.

NOTE: If your classroom has no geocoding credits, skip Path B and discuss the workflow verbally; use Path A for all analysis labs.

6.3 Quick QA

- Open attribute table → Select Layer By Attribute → ARPU_USD >= 25; how many high-value customers?
- Select CHURN_RISK >= 0.55; compare spatial pattern visually against CellSites.

CHECKPOINT: Matched subscribers appear as points; high-ARPU and high-churn selections work.

7. Lab 2 — Density Mapping & KPI Joins

7.1 Kernel Density (subscriber concentration)

- Ensure map/analysis is in a projected meter-based system for density (Project Subscribers to UTM if still in WGS84 degrees).
- Analysis → Tools → Kernel Density
- Input point features = Subscribers (projected)
- Population field = NONE (counts) — optional second run with Population field = ARPU_USD for revenue-weighted density
- Output cell size = 50 (meters) or 100 if slow

- Search radius = 400 to 800 m (try 500 m first)
- Output = KD_Subscribers and KD_ARPU
- Symbology = Stretch or Classify (quantile); apply a heat-map color ramp (light → dark red)

TIP: In Pro you can also use Symbology → Heat Map on the point layer for a quick visual; Kernel Density is better for reuse in overlay models.

7.2 Attribute Join — Zone KPI table

- Add Zone_KPI_Table.csv to the map/project
- Join Field or Add Join: MarketingZones + Zone_KPI_Table on DIST_ID = DIST_ID
- If using Add Join, export to MarketingZones_KPI to persist fields
- Symbology on AVG_ARPU (graduated colors) and create a second copy for AVG_CHURN_RISK
- Label DIST_NAME and SUBSCRIBERS

7.3 Spatial Join — customers to zones (verify consistency)

- Spatial Join: Target = MarketingZones; Join = Subscribers; Operation = One to One; Match = Intersect
- Field mapping / merge rules: ARPU_USD → Mean; CHURN_RISK → Mean; CUST_ID → Count
- Output = Zones_SpatialJoin_Stats
- Compare Mean ARPU from spatial join vs AVG_ARPU from Zone_KPI_Table (should be very close).

CHECKPOINT: You have a heat/density surface and zones colored by ARPU and churn.

8. Lab 3 — Hot Spots & Outliers (High-value / At-risk)

Objective: Move from “eyeballing clusters” to statistically significant segments.

8.1 Prepare analysis points

- For speed, you may Aggregate customers into zones and run Gi* on zones; for finer detail run on points.
- Recommended classroom path: run Hot Spot on MarketingZones_KPI using field AVG_ARPU, then again on AVG_CHURN_RISK.
- Optional advanced path: run Optimized Hot Spot Analysis on Subscriber points using ARPU_USD.

8.2 Hot Spot Analysis (Getis-Ord Gi*)

- Spatial Statistics Tools → Mapping Clusters → Hot Spot Analysis (Getis-Ord Gi*)
- Input Feature Class = MarketingZones_KPI
- Input Field = AVG_ARPU
- Conceptualization = Contiguity edges corners (polygon) OR Fixed Distance Band if using points
- Output = HS_ARPU

- Repeat for Input Field = AVG_CHURN_RISK → HS_CHURN
- Interpret Gi_Bin / Gi_Conf levels: +3 hot spot 99%, +2 95%, +1 90%; negative = cold spots

Business reading:

- ARPU hot spots = priority upsell / premium care / fiber cross-sell zones.
- Churn hot spots = retention campaigns, QoS investigation with Day 2 coverage layers.
- ARPU cold + Churn hot = urgent win-back / quality fix areas.

8.3 Cluster & Outlier Analysis (Anselin Local Moran's I)

- Spatial Statistics Tools → Mapping Clusters → Cluster and Outlier Analysis (Anselin Local Moran's I)
- Input = MarketingZones_KPI; Field = AVG_ARPU; Output = CO_ARPU
- Review COType values:

COType	Meaning	Geo-marketing action
HH	High value surrounded by high	Defend & upsell
LL	Low value surrounded by low	Cost-efficient acquisition / coverage first
HL	High outlier among low	Protect VIP pocket
LH	Low outlier among high	Investigate leakage / competitor win
Not Significant	No strong local pattern	Monitor

- Optional: repeat Cluster & Outlier on CHURN_RISK.

CHECKPOINT: HS_ARPU / HS_CHURN and CO_ARPU layers created; you can name top 3 HH and top 3 churn hot zones.

9. Lab 4 — Market Expansion Suitability

Objective: Rank where MetroTel should expand sales / sites using multi-criteria overlay.

9.1 Criteria & weights (from Suitability_Weights.csv)

Criterion	Weight %	Reclass logic (1=low ... 9=high suitability)
Population / market size	25	Higher population → higher score
Average ARPU	20	Higher ARPU → higher score
Churn risk (inverse)	15	Lower churn → higher score for expansion offers
Throughput / QoS	15	Higher throughput → higher score

Competitor pressure (inverse)	10	Lower competitor density → higher score
Affluence index	15	Higher affluence → higher score

NOTE: Weights must sum to 100%. Instructors may change weights for a “what-if” discussion.

9.2 Build raster criteria (Spatial Analyst path)

- Polygon to Raster MarketingZones_KPI → rasters for POPULATION, AVG_ARPU, AVG_CHURN_RISK, AVG_THROUGHPUT, AFFLUENCE (from zones) at cell size 50–100 m.
- Kernel Density on CompetitorSites → CompDensity; Remap so low density = high suitability.
- Reclassify each raster to a common 1–9 scale (Reclassify tool). For churn & competitor, reverse the scale.
- Weighted Overlay: add reclassified rasters with weights above; evaluation scale 1–9; output = Suitability_MarketExpansion
- Alternative in newer Pro: Suitability Modeler with same criteria/weights (transform → weight → locate).

9.3 Zone ranking without rasters (lighter classroom path)

- In MarketingZones_KPI attribute table, Add Fields: SCORE_POP, SCORE_ARPU, SCORE_CHURN_INV, SCORE_THP, SCORE_AFF, SCORE_COMP, SUITABILITY
- Use Field Calculator / Classify ranks 1–9 within each field (or manual reclass using quantiles).
- $SUITABILITY = 0.25 * SCORE_POP + 0.20 * SCORE_ARPU + 0.15 * SCORE_CHURN_INV + 0.15 * SCORE_THP + 0.10 * SCORE_COMP + 0.15 * SCORE_AFF$
- Sort SUITABILITY descending → top 3 districts are expansion priorities.
- Spatial Join candidate CellSites (STATUS=Candidate) to zones; see which candidates fall in top-ranked districts.

9.4 Map product

- Create layout 1: Subscriber Density + CellSites
- Create layout 2: ARPU Hot Spots + Churn Hot Spots (side by side or two frames)
- Create layout 3: Market Suitability ranking + Candidate sites
- Include title, legend, north arrow, scale, data source note: “Synthetic MetroTel Day 3 training data”
- Export PDF: Day3_GeoMarketing_Layouts.pdf and optionally Share as Map Package (.mpkx)

CHECKPOINT: Suitability scores exist; at least one layout exported; top expansion zones identified.

10. Expected Outputs Checklist

#	Deliverable	Done?
1	MetroTel_Day3_GeoMarketing.aprx + MetroTel_Day3.gdb	
2	Subscribers point feature class (from XY and/or geocode)	

3	KD_Subscribers and/or heat-map symbology	
4	MarketingZones_KPI (joined ARPU/churn/QoS)	
5	HS_ARPU and HS_CHURN hot spot results	
6	CO_ARPU cluster/outlier result	
7	Suitability_MarketExpansion OR SUITABILITY field ranking	
8	One-page summary: top 3 high-value zones, top 3 at-risk zones, top 3 expansion zones	
9	Exported map layout PDF	

11. Instructor Answer Key — Synthetic Sample Findings

Exact ranks can vary slightly with projection/radius choices; these are typical results from the packaged seed=42 dataset:

- Largest subscriber counts usually concentrate in Central Business District, North Residential, South Corridor.
- Enterprise / high ARPU often stronger in CBD / Tech Park / University Belt.
- Higher churn risk often appears where sites are sparse relative to demand (e.g., expansion/peri-urban districts).
- Candidate sites CAND-01..04 should be evaluated against suitability ranks — not all candidates are equally good.
- Competitor clusters reduce suitability if the inverse-competition weight is applied.

12. Troubleshooting

Problem	Likely cause	Fix
XY points land in wrong country	Lon/Lat swapped or wrong CRS	X=LON, Y=LAT; GCS WGS84 then project
Kernel Density tool fails / tiny output	Layer in geographic degrees	Project points to UTM/meters first
Geocode uses many credits / fails	No locator / offline class	Use Subscribers_with_XY Path A
Join produces nulls	DIST_ID mismatch / type mismatch	Check text vs text; trim spaces
Hot Spot all “Not Significant”	Too few polygons / wrong conceptualization	Use 12 zones with contiguity; or use point Optimized Hot Spot
Weighted Overlay error	Criteria not integer 1–9	Reclassify all inputs to same scale

13. Quick Quiz (5 questions)

- What is the difference between Kernel Density and Hot Spot Analysis?
- Which join type do you use to attach Zone_KPI_Table.csv to MarketingZones by DIST_ID?

- If a zone is HH for ARPU and a hot spot for churn, what business action makes sense?
- Why reverse-score competitor density in a market-expansion suitability model?
- Name two real open datasets you could use instead of synthetic population/zones.

Suggested answers:

- 1) Density = smooth concentration surface; Hot Spot = statistically significant clusters vs random.
- 2) Attribute join / Join Field on DIST_ID (not spatial join).
- 3) High-value but unstable — priority retention + QoS/coverage check.
- 4) High competition lowers expansion attractiveness (unless strategy is head-to-head).
- 5) e.g., HDX COD boundaries + WorldPop population (or OSM + regulator/OpenCellID context).

14. Bridge to Day 4

Day 4 moves from customer geography to physical network design: FTTH/transmission geometric networks, microwave LOS, terrain profiles, and viewsheds. Bring forward today's priority expansion zones — Day 4 can test whether fiber/microwave options can serve them.

15. Version History

Version	Changes
v1	Draft outline of Day 3 geo-marketing topics
v2	Added lab sequence and KPI join examples
v3	Full manual with synthetic MetroTel datasets, real open-data options, detailed ArcGIS Pro steps for Geocoding/XY, Kernel Density, Joins, Hot Spot, Cluster & Outlier, Suitability/Weighted Overlay, checkpoints, answer key, quiz

— End of Day 3 Manual (Version 3.0) —